

Cyclistic Bike-Share Case Study: Increasing Annual Memberships

Executive Summary

This case study analyzes **3.97 million valid trip records** from August 2025 through July 2026 for Cyclistic (Chicago's Divvy bike-share system)[cite: 1]. The primary goal is to uncover behavioral differences between casual riders and annual members to design targeted, high-ROI marketing strategies that convert casual riders into annual members[cite: 1].

Business Problem (Ask)

- **Objective:** Increase annual memberships by identifying opportunities to convert casual riders into subscribers[cite: 1].
- **Key Business Questions:**
 1. How do trip volumes and usage patterns differ between casual riders and members?
 2. When (months, days, hours) do casual riders use Cyclistic bikes most frequently?
 3. How do the two groups differ in trip duration and bike type choices?
 4. Which high-volume stations have a high casual rider concentration for targeted marketing campaigns?
- **Key Stakeholders:** Lily Moreno (Director of Marketing), Marketing Analytics Team, and Cyclistic Executive Team.

Data Source & Data Preparation (Prepare & Process)

- **Data Source:** Publicly available historical trip data provided by Motivate International Inc. / Divvy (stored in AWS).
- **Initial Dataset:** 12 monthly CSV files covering August 2025 – July 2026, totaling **6,073,968 raw trip records**.
- **Data Cleaning & Integrity (SQL & Excel):**
 - Removed **1,273,200 records** missing start_station_id and **752,047 records** missing end_station_id.
 - Eliminated **20 duplicate records** and **19 invalid trips** where start time occurred after end time.
 - Fixed file naming discrepancies (e.g., corrected 202501-divvy-tripdata containing January 2026 data).

- **Feature Engineering & Outlier Removal:**
 - Derived day_of_week, ride_duration_seconds, and ride_speed_kmh.
 - Removed 173 zero-duration rides and 43,096 rides under 1 minute.
 - Excluded implausible speed outliers (> 35 km/h).
- **Final Analysis Dataset: 3,969,413 validated trip records.**

Key Findings & Insights (Analyze)

1. Trip Volume & Seasonal Trends

- **Rider Distribution:** Members generated **65.12%** (2.58M) of total trips, while casual riders accounted for **34.88%** (1.38M).
- **Seasonality:** Casual rider shares surges in warmer months (peaking at **42.9%** in August) and drops significantly during winter (down to **17.05%** in December).

2. Weekly & Hourly Usage Patterns

- **Weekend Surge:** Casual usage shifts heavily toward weekends—reaching **48.42%** on Sundays and **46.10%** on Saturdays—whereas members dominate weekday commuting (up to **73.1%** on Tuesdays).
- **Time of Day:** Casual riders make up the majority of trips during late night/overnight hours (midnight to 4:00 AM, peaking at **61.73%** at 2:00 AM).

3. Trip Duration & Bike Preferences

- **Trip Duration:** Casual riders favor longer rides; they represent **83.5%** of trips lasting 1–2 hours and **82.5%** of trips lasting 2–4 hours. Members dominate shorter trips (< 30 minutes).
- **Bike Preference:** Electric bikes represent **37.27%** of casual trips compared to **32.23%** of classic bike trips, demonstrating a slight preference for electric bikes among casual riders.

Interactive Station Analysis (Share)

 **Interactive Dashboard:** [Explore the Station-Level Analysis on Tableau Public](#) (Insert your actual link)

Using Tableau, stations were filtered by two key metrics: **minimum 5,000 total trips** and **≥ 40% casual rider proportion**.

- **Target Identified: 42 high-volume stations** met the criteria[cite: 1].
- **Geographic Focus:** Highly concentrated along the Lake Michigan lakefront and central Chicago areas (Downtown, South Loop, River North, Lincoln Park, and Lakeview).

Strategic Recommendations (Act)

1. **Seasonal & Weekend Campaigns:** Launch targeted membership promotions in late spring/summer and execute weekend-specific digital ad campaigns when casual rider activity peaks.
2. **Geofenced & Station-Level Targeting:** Focus physical signage, QR codes, and targeted mobile ads at the **42 identified high-volume lakefront and downtown stations**.
3. **Value Messaging on Trip Duration:** Frame membership messaging around long-ride value savings for users frequently taking trips over 45 minutes.
4. **A/B Testing & Evaluation:** Test varied messaging offers (e.g., weekend pass conversion credits vs. annual membership discounts) and track conversion rates, acquisition costs, and retention rates.